

Introduction to Constructing Insurable Risk Portfolios

Colombia Workshop

Edward (Jed) Frees, University of Wisconsin - Madison

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The Workshop

- ① Participants learn how to construct an optimal investment portfolio
- ② Participants will work with code to quantify risk measures and risk retention functions. You also learn to optimize basic portfolios.
- ③ Two Case Studies
 - I will give overviews of the ANU and Wisconsin cases
 - During workshop time, participants will work on one of these cases (your choice)
- ④ Participants will explore Florida flood insurance data

Outline of Presentation

- The Topic
- Algorithmic Framework - Risk Retention
- Investment Portfolios

Constructing Insurable Risk Portfolios

- Actuaries routinely advise risk managers on a host of different issues
 - These issues include pricing, reserving, underwriting, claims management, and capital allocation
 - Typically, we think about an insurer (or reinsurer) as the recipient of this advice
- Instead, I will consider the recipient of the advice to be a generic person, firm, or organization
 - This recipient is interested in advice on how much risk (in their business or lives) should be transferred

Let us start by unpacking the title “*Constructing Insurable Risk Portfolios*”

People, Firms, Organizations Face Multiple Risks

Table. **Top Insurable Risks Facing Firms**

Risk	Risk
4. <i>Business interruption</i>	44. Directors and Officers liability
19. Counter-party credit risk	47. Fraud
21. <i>Property damage</i>	52. Theft
22. Environmental risk	55. Terrorism sabotage
23. Weather natural disasters	56. Safety and Pharmacovigilance
24. <i>Third party liability</i>	61. Harassment discrimination
28. Injury to workers	66. Kidnap and ransom
40. Product recall	67. Extortion

A risk is **insurable** if it potentially can be transferred to another party for a fee.

Australian National University (ANU) Case

Class of Insurance	Insurer	Limit / Deductible	Premium
Property	London Syndicate and Others	5,000,000	23,564,759
General and (G & P)	Newline, Liberty, QBE, Chubb,	100,000	448,550
Products Liability	CGU and Zurich		
Cyber	London	250,000	75,721
Crime	AIG	100,000	100,000
Employment	AIG	100,000	84,000
Practices Liability			
Expatriate	Chubb		11,676
Group Personal	Chubb	Various	104,920
Accident			
Corporate Travel	Chubb	Various	75,000
Professional	Newline	100,000	130,000
Indemnity			
Medical Malpractice	Newline	100,000	
Clinical Trial	Newline	2,500	
Statutory Liability	Berkley Insurance Australia (SUA)	\$1,000 / \$15,000	8,360
Motor Vehicle	Vero	1,000	84,700
Marine Cargo	Richard Oliver (QBE)	5,000	6,127
Marine Hull	Richard Oliver (QBE)	150	11,552
TOTAL			24,407,255

Australian National University (ANU) Case

- A typical large firm that needs protection from several risks
 - To illustrate its size, in 2020 it enjoyed an asset base of about 4.6 billion AUD
- To illustrate its complexity, its financial statements summarize many risks including property damage, general liability, cyber security, and so forth.
 - There are 15 risks in total, the “usual suspects.”
 - It pays about 25 million AUD per year in insurance premiums.
- In 2020 ANU suffered a major property loss (250 million AUD) meaning that its risk appetite has been subject to renewed scrutiny
 - Need models that allow for long-tail distributions that permit very high values of outcomes with not insignificant probabilities.

Abstracting the Problem

- Think about each of the 15 ($= p$) risks as X_j , $j = 1, \dots, p$
- For each risk, the risk owner (ANU)
 - retains some risk, $g_j(X_j)$
 - and cedes risk to another party $X_j - g_j(X_j)$

Abstracting the Problem

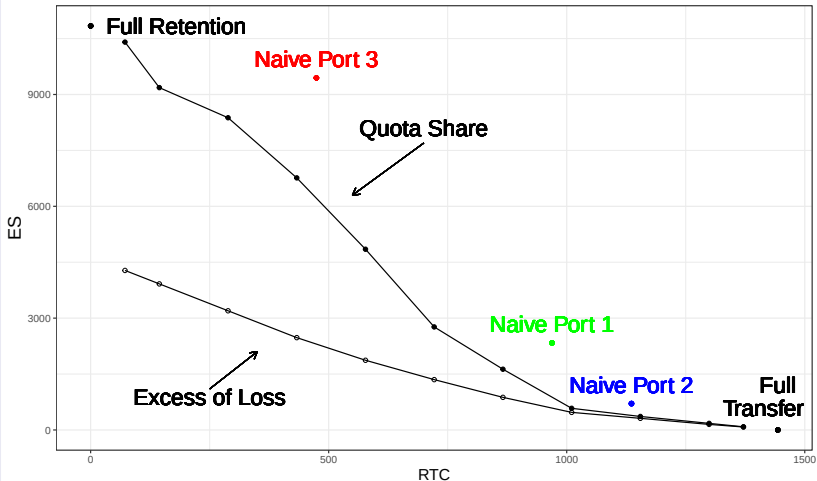
- Think about each of the 15 ($= p$) risks as X_j , $j = 1, \dots, p$
- For each risk, the risk owner (ANU)
 - retains some risk, $g_j(X_j)$
 - and cedes risk to another party $X_j - g_j(X_j)$
- There is a cost / price of ceding these risks, denoted as RTC (risk transfer cost)
 - For example, use expectations to start
$$RTC = E \left\{ \sum_{j=1}^p X_j - g_j(X_j) \right\}$$
- There is uncertainty associated with the retained risks.
 - The total retained risk is $g(\mathbf{X}) = \sum_{j=1}^p g_j(X_j)$
 - Capture this uncertainty with a risk measure, $RM[g(\mathbf{X})]$

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 - Capture this uncertainty with a risk measure, $RM[g(\mathbf{X})]$
- Naturally, there is a trade-off between RTC and $RM[g(\mathbf{X})]$
- Our goal is to quantify this trade-off for different types of retention plans

Comparison of Efficient Frontiers

Also shown are three naive portfolios that do not lie on either frontier.



Four Case Studies

In this workshop, we will discuss four different case studies:

- We begin with classical (investments) portfolio decisions inspired by the Markowitz framework.
- We consider risk retention for a large commercial enterprise (ANU - Australian National University)
- Wisconsin Property Fund - similar to data from a commercial insurer selling, for example, business owners policies
 - Exposure decisions for an insurer's portfolio of small commercial policies
- Florida flood risk data will allow us to feature spatial aspects of risk retention

Algorithmic Framework - Risk Retention

- Think about p risks (such as 15), denoted as $X_1, \dots, X_p$
- Each risk is subject to an insurance contract that specifies:
 - $g_j(X_j; \theta)$ - the amount retained
 - $X_j - g_j(X_j; \theta)$ - the amount transferred to another party

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These contracts can be complex, so we start at the foundations:

- Suppose that $g(x; \theta = c) = c \cdot x$, a *coinsurance* or *quota-share*
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 - For example, your insurer pays 80% of your prescription drugs expenses, you pay 20%
- Suppose that $g(x; \theta = u) = \min(x, u)$, known as *deductible* or *excess of loss*
 - For example, you retain claims from your auto accident up to 250, the insurer pays the excess
 - Mildenhall and Major (2022) refer to this type of contract as the “basic building block of insurance.”
- I refer to c, u, θ as **risk retention** parameters

Algorithmic Framework - Risk Retention

- When selecting risk retention parameters, there are trade-offs:
 - the uncertainty of the amount of risk retained and
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Algorithmic Framework - Risk Retention

- When selecting risk retention parameters, there are trade-offs:
 - the uncertainty of the amount of risk retained and
 - the cost of purchasing risk transfer (insurance).
- To quantify this, define the retained risk

$$g(\mathbf{X}; \theta) = \sum_{j=1}^p g_j(X_j; \theta)$$

- Denote the uncertainty as $RM[g(\mathbf{X}; \theta)]$.
 - For the RM (risk measure), variance is a possibility. I typically use expected shortfall.
- The risk transferred is $X_1 + \dots + X_p - g(\mathbf{X}; \theta)$.
 - The cost of the transfer is denoted as $RTC(\theta)$
 - It is evaluated using an insurance pricing principle, starting with an expectation (augmented for administrative costs)

Risk Retention Optimization Problem

$$\begin{array}{lll} \text{minimize}_{\theta} & RM[g(\mathbf{X}; \theta)] & \text{uncertainty} \\ \text{subject to} & RTC(\theta) \leq RTC_{max} & \text{budget restriction} \\ & \theta_j \geq 0 & \text{parameter restrictions} \end{array}$$

Risk Retention Optimization Problem

minimize θ	$RM[g(\mathbf{X}; \theta)]$	<i>uncertainty</i>
subject to	$RTC(\theta) \leq RTC_{max}$	<i>budget restriction</i>
	$\theta_j \geq 0$	<i>parameter restrictions</i>

- A constrained optimization problem - easy to interpret
 - RTC_{max} is a user provided budget
 - Could easily add other parameter restrictions (e.g., $\theta \leq 1$ for coinsurance)
- Behind the scene, computation is complex
 - Not a convex problem (like machine learning)
 - Requires simulation techniques to compute

Portfolio of Investments

- For illustration, consider a portfolio of $p = 8$ asset risks from the insurance sector. With ticker symbols in parens (*)*):
 - United Health (UNH) - *USA*
 - AIA Group Limited (1299.HK) - *Hong Kong*
 - Ping An Insurance (Group) Company of China, Ltd. (2318.HK) - *China*
 - Chubb (CB) - *Switzerland*
 - Allianz (ALV.DE) - *Germany*
 - Aon (AON) - *UK*
 - AXA (CS.PA) - *France*
 - Tokio Marine (8766.T) - *Japan*.
- One buys a certain amount of each risky asset
- The collection of risky assets owned is the **portfolio**

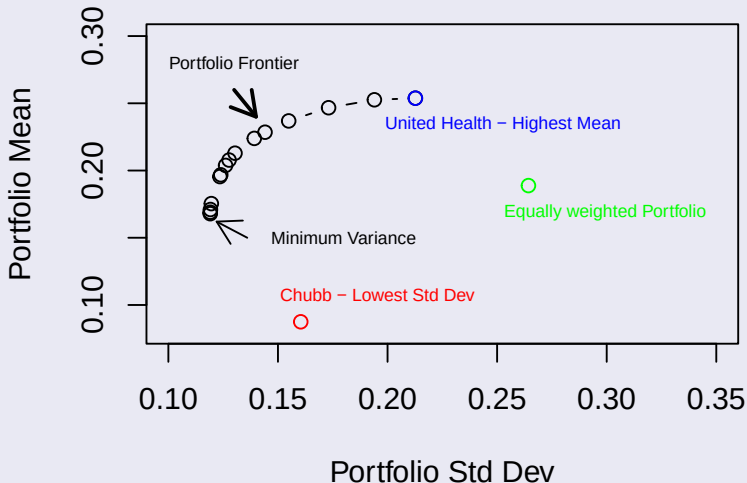
Comparing Investment and Insurable Risk Problems

For interpretation, think about the well-known Markowitz (investment) portfolio construction.

	Investment	Insurable Risk
Risk Retention Parameters	$\theta_1, \dots, \theta_p$	$\theta_1, \dots, \theta_p$
Retained Risk	$S_{Inv}(\mathbf{X}, \theta) = \theta_1 X_1 + \dots + \theta_p X_p$	$g(\mathbf{X}, \theta) = g_1(\theta_1, X_1) + \dots + g_p(\theta_p, X_p)$
Objective Function minimize $_{\theta}$	$Var[S_{Inv}(\mathbf{X}, \theta)]$	$RM[g(\mathbf{X}, \theta)]$
Budget Constraint subject to	$E[S_{Inv}(\mathbf{X}, \theta)] \geq Req_0$	$RTC(\theta) \leq RTC_{max}$

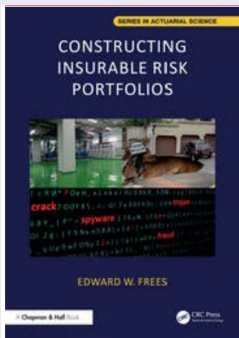
Constructing an Optimal Portfolio

Employing techniques due to Markowitz (1952), one can create a *frontier* of optimal portfolios.



Concluding Remark

- You can get an overview of this topic in my [2025 book](#).



Thank you for your time and attention.